Tharun V Puthanveettil

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— Senior Robotics Researcher with 5+ years of experience building perception, manipulation, and learning systems for real robots.
— Focused on human-in-the-loop vision, control, and data pipelines for closed-loop robotic autonomy.
— Published researcher with 100+ citations and 3 patents (filed/pending).

Education

University of Maryland - CP, USA | **Master of Engineering in Robotics (GPA: 3.9/4.0)** Aug. 2022 – May 2024
Vellore Institute of Technology, India | **B.Tech. in Electronics and Communication (GPA: 8.59/10.0)** Jul. 2014 – Apr. 2018

Experience

Robotics Engineer | **Deformable Object Manipulation, Robot Learning, Multimodal Perception** Jun. 2024 – Present
General Motors R&D *Michigan, USA*

- Architected a modular **ROS 2 robot-learning platform** unifying bimanual teleoperation, task-conditioned perception, physics simulation, **MoveIt 2** planning, multimodal data capture, and learned-policy evaluation.
- Built task-conditioned detection, segmentation, tracking, and pose-servoing services on **Foundation Models**; cut perception latency ∼35% and sustained 30–40 FPS via embedding reuse and temporal mask propagation.
- Developed bimanual manipulation of **deformable linear objects** (cable/harness assemblies) using impedance control and force–torque feedback; bounded elastic tension and measured a 2× failure increase without tension regulation.
- Constructed high-fidelity **digital twins** with domain randomization and behavior-tree-driven synthetic data generation to quantify **Sim2Real** degradation under sensing, dynamics, and actuation shift.
- Designed a config-driven **teleoperation framework** spanning six control modes, five input modalities, live switching, and N-arm scaling; validated on a dual-arm industrial cell with VR and force-controlled grippers.
- Established an evidence-governed **data-to-policy pipeline**: synchronized multimodal recording, automated QC, semantic review, frozen splits, dataset conversion, shadow inference, and staged rollout gates.
- Improved object-detection robustness by **20–30%** through hard-negative sampling, structured distractor generation, and automated synthetic-data augmentation.
- Led a large-scale **Imitation Learning** study benchmarking action-chunking transformer against **Vision–Language–Action** policies; designed confirmation-gated skill orchestration with typed human-in-the-loop takeover.

Researcher (AI & Robotics) | **Multi-Modal Networks, Data Security, Action Recognition** Feb 2023 – Mar 2024
CATT Labs *Maryland, USA*

- Developed *Multi-Modal Anomaly Detectors* to report behaviors that deviate from the robot's normal baseline nature.
- Created a one-stop log retrieval service using **Large Language Models** to improve accessibility for non-technical users.
- Formulated **metrics** to measure, compare, and parameterize the safety of robots and cyber-physical systems.

Track Anything Rapter (TAR) | *Python, PyTorch, OpenCV, ROS2* [github] [preprint]

- Realized a UAV system to achieve **one-shot object tracking** using **Foundation Models** from multi-modal queries.
- Implemented a custom **ROS2 Gazebo** simulation pipeline for validation of tracking algorithms.
- Developed a **proportional-based visual servoing controller** leveraging **PX4 Autopilot** for closed-loop object tracking.
- Transitioned to a **ROS2**-based control system utilizing **MAVROS**, bypassing a dependency on **MAVSDK**.
- Achieved a mean **Dynamic Time Warping (DTW)** score of 1.68 for UAV trajectory against **Vicon** ground truth.

Fall Detection using Human Pose Estimation | *Python, PyTorch, OpenCV* [github1] [github2][preprint]

- Crafted a **Human Pose Estimation**-keypoint-based fall detection model; identified optimal framerate for fall events.
- Quantified the potency of each **pose keypoint** and the relevance of **camera angle** in improving accuracy.
- Custom **GCN + Transformer** model achieved 95% (avg.) accuracy on both UR & NTU datasets.
- Devised a pipeline for multi-view-based pose aggregation for fall recognition in **partial/no occlusion** scenarios.
- **GCN+Transformer** multi-view aggregation model achieved 96% accuracy for multi-class action recognition.

Researcher (AI & Robotics) | **Human-Robot Interaction, Computer Vision, Precision Manipulation** Jun. 2021 – Jan. 2023
Indian Institute of Science & Perception and Robotics Group, UMD *Bangalore, India & Maryland, USA*

- Implemented an *Imitation Learning-based policy* to instruct robot actions from human demonstration, speech & gestures.
- Constructed a **U-Net**-based **Monocular Depth Estimation** model on the NYU dataset with 94% accuracy (Top 4 @ IROS21).

Autonomous Weeding Robot | *Python, Matlab, OpenCV, TensorFlow, PyTorch, ROS* [video] [github]

- Engineered a **robotic manipulator**-based plucking robot to reduce pesticide usage in indoor farms by up to 80%. [Report - Link]
- Created a **YoloR** object detector trained on augmented synthetic data for weed localization with 98% accuracy.
- Implemented an IK solver based on supervised **Behavioral Cloning** for precise manipulation.
- Developed a precision weeding gripper with adaptive vision-based control for minimal non-weed contact. [Patent - Link]
- Designed an **adaptive PID controller** to enable weeding while minimizing disturbance to the plant bed.

Project Engineer | **HCI, Photogrammetry, Recommendation Engine, Pattern Recognition** Jul. 2018 – May 2021
CTO Office, Wipro Digital *Bangalore, India*

- Led as the **AI Project Lead** at the Innovation team, orchestrating the quarterly project lineup.
- Developed a **2D CNN**-based tool tracking system using Leap Motion controller, enhancing hardware assembly pipelines.
- Engineered a mobile-centric insurance solution with a **Transfer Learning**-based car model detector (96% acc.) in **TFLite**.
- Implemented **Generative AI** and learning-based algorithms for *my Style*, one of Wipro's top retail POCs (2020).

my Style | *Python, OpenCV, TensorFlow, PyTorch, MySQL, Flask* [link1] [link2] [video]

- End-to-end AI-powered shopping app to reduce trial room drops and improve scope for apparel customization.
- Implemented body measurement extraction via **2D photogrammetry** & **Human Pose Estimation**.
- Developed a **RASA**-based **chatbot** for personalized dress recommendations.
- Built a **GAN**-based product customization model for dynamic apparel styling (82% higher customer traction).
- Engineered a supervised **Fit Analyzer** model to evaluate apparel fit as a 'Fit%' metric.

Publications/Patents/Posters

[1] Geometrical Feature based Precision Farming using Hybrid IK Solver and Flower Carpet Planner for Commercial Greenhouse Environments. ICSR, 2025. [Link]
[2] ASTROD: Multimodal Anomaly Detection for Autonomous CPS Empowering Real-World Evaluation. GameSec 2024. [Link]
[3] A Data-Driven Approach for Securing Autonomous Cyber-Physical Systems. CDC 2025. [Manuscript]
[4] Application of Mobile Collaborative Robot using Deep Learning in Precision Weed Control. Elsevier 2021. [Manuscript]
[5] Event-based Dynamic Obstacle Avoidance in Outdoor Environments. IROS 2021. [Poster]
[6] A univariate data analysis approach for rainfall forecasting. ICCIS 2020. [Link]
[7] Prediction of rainfall using data mining techniques. ICICCT 2018. [Link]
[8] A comparative study of various classification techniques to determine water quality. ICICCT 2018. [Link]
[9] Wall climbing robot using soft robotics. ICPCSI 2017. [Link]

Technical Stack

Programming: Python, C/C++, SQL (MySQL), JavaScript, HTML/CSS, Matlab, LabVIEW, Docker
AI/ML/CV/Data: TensorFlow, PyTorch, OpenCV, Keras, scikit-learn, Tableau, LangChain, Pydantic, OpenAI, NLTK
Embedded/IoT/Robotics/Sim: ROS, ROS2, MQTT, HTTP, REST, Sockets, Soft Robotics, Unity, Flask, Gazebo, RViz
Robotic Platforms: Turtlebot 2/3, Delta, OpenManipulator-X, Husky, M500 Drone, UR5/UR5e, UR10
S/W Development: Agile, Jira, Git/GitLab/Bitbucket, Pair Programming, TDD, UML, SCM

Additional Projects

AI | Computer Vision | Machine Learning

Image Segmentation using SLIC | *Python, OpenCV, TensorFlow, HTML, CSS, Angular.js, Unity* [link] — Nov. 2023 – Dec. 2023
- Utilized **Simple Linear Iterative Clustering (SLIC)** to generate superpixels for a custom segmentation dataset.
- Implemented a CNN-based deep feature extractor for superpixel-based image segmentation.
- Achieved 20% improved segmentation performance over **k-NN**-based segmentation.

Image Outpainting | *Python, OpenCV, TensorFlow, HTML, CSS, Angular.js, Unity* [link] — Nov. 2023 – Dec. 2023
- Implemented Image Outpainting using **Implicit Neural Representation**.
- Observed $\sim$3-point lower PSNR using **Positional Encoding** add-on vs. FFN-only baseline.

AI Cricket Coach | *Python, OpenCV, TensorFlow, HTML, CSS, Angular.js, Unity* [link] — Jul. 2020 – Present
- Classifies bowl types from video feeds; provides coaching assistance in a **WebVR** environment.
- Created a **Body Pose Estimation** plugin for batting pose feedback.
- Custom **3D CNN** bowl classifier (92% accuracy) trained on synthetic VR data.

Virtual Try-On | *Python, OpenCV, TensorFlow, PyTorch, Flask* [link] — Jan. 2021 – Mar. 2021
- AI solution to reduce trial room drops and provide a style quotient evaluator.
- Built a **GAN**-based texture generator rendered onto **3D AR** apparel models.
- Developed an **Aesthetic Quality Assessment Model (AQAM)** and an **Explainable AI** plugin.

Robotics

Maze Runner | *C++, ROS2, Gazebo, RViz* [link1] [link2] — Nov. 2023 – Dec. 2023
- Autonomous maze navigation using dynamic cues from **ArUco** markers in Gazebo.
- Implemented a **ROS2 Action-Client** that uses **Nav2** for visual feed–based waypoint navigation.

Leonardo - Autonomous Retrieval UGV | *Python, Arduino, OpenCV* [video] — Feb. 2023 – May 2023
- Constructed a Barron robot with IMU, encoder, RPi camera, and a servo-based arm.
- **Visual Servoing**-based low-level controller for object pick-and-place with onboard arm.
- Rapid localization via **sensor fusion** of IMU, encoder, & range sensor in confined zones.

Auto Platoon | *Python, OpenCV, PyTorch* [video] [link] — Apr. 2023 – May 2023
- Bio-inspired multi-agent leader–follower system for modular & energy-efficient transport.
- **YOLOv7**+Kalman-based agent tracking with dynamic obstacle avoidance.
- Socket-based connected-vehicle communication and **Visual Servoing**-based motion planning.

ACO–RRT* — Bio-Inspired Path Planning | *Python, NumPy* [link] — May 2023 – Jun. 2023
- Implemented a bio-inspired planner that explores random trees via ant foraging behavior.
- Improved traditional **RRT*** sampling with **Ant Colony Optimization**.
- Achieved 1.4$\times$ and 3.54$\times$ faster convergence (vs. **RRT***) for first and ideal paths, respectively.

Volunteering Experience

Electrical Programming Head Coach | **Robotics, Software Development, Computer Vision** — Jun. 2024 – Present
The Irrational Engineers — *Maryland, USA*
- Mentoring diverse high school students in building competitive robots for **FIRST Robotics**, focusing on software for **Controls** and **Computer Vision**.

GCN - Graph Convolution Network; GAN - Generative Adversarial Network; TSA - Time Series Analysis; SCM - Software Configuration Management; UGV - Unmanned Ground Vehicle.